

Graph Puzzles III.1:

A proof of Sabidussi's compatibility conjecture

Nikolay Ulyanov*

*ulyanick@gmail.com

ABSTRACT: We prove Sabidussi's compatibility conjecture. Let G be a finite connected multigraph in which every vertex has even degree and the minimum degree is at least four, and let T be a closed trail that traverses every edge exactly once. The edges of G can be partitioned into circuits—nonempty connected subgraphs in which every vertex has degree two—so that no circuit contains the two edges used consecutively at any passage of T . In fact, the edges can be four-coloured so that every such pair receives two different colours and the subgraph formed by the edges of each colour has even degree at every vertex.

1 Introduction

Throughout, all graphs are finite multigraphs with nonempty vertex sets; loops and parallel edges are allowed. We write $V(G)$ and $E(G)$ for the vertex and edge sets of a graph G , and $|X|$ for the cardinality of a finite set X . Every edge has two *incidences*, or *ends*, each assigned to a vertex. An edge is a loop when its two ends are assigned to the same vertex; distinct non-loop edges are parallel when their ends are assigned to the same two vertices. The degree $\deg_G(v)$ counts edge incidences at v , so a loop contributes two. For a nonempty graph, the minimum degree is

$$\delta(G) = \min_{v \in V(G)} \deg_G(v).$$

A graph is *even* when every vertex has even degree, and it is *Eulerian* when it is connected and even.

For an integer $k \geq 0$, a *trail of length k* is an alternating sequence

$$v_0, e_1, v_1, e_2, \dots, e_k, v_k$$

in which the two ends of e_i are assigned to v_{i-1} and v_i , for $1 \leq i \leq k$, and the edges $e_1, \dots, e_k$ are distinct. It is closed when $v_0 = v_k$. An *Euler tour* is a closed trail containing every edge exactly once, regarded up to cyclic shift. A *circuit* is a nonempty connected subgraph in which every vertex has degree two. A *circuit decomposition* of G is a family of pairwise edge-disjoint circuits whose edge sets partition $E(G)$.

A *transition* at a vertex v is an unordered pair of distinct edge incidences at v . A circuit contains the transition when it contains each underlying edge represented by the pair, and is *compatible* with the transition otherwise. A *transition system* is, at every vertex, a

partition of its incident edge incidences into transitions. An Euler tour induces such a system: at every vertex visit, pair the incidence by which the tour arrives with the incidence by which it next departs.

Sabidussi's conjecture concerns the transition system induced by a single Euler tour. We prove it in the following form.

Theorem 1.1 (Sabidussi). *Let G be a finite Eulerian multigraph with $\delta(G) \geq 4$, and let T be an Euler tour of G . There is a circuit decomposition $\mathcal{C}$ of G such that no $C \in \mathcal{C}$ contains both underlying edges of any transition of T .*

The reduction from the graph to a cyclic-word colouring problem begins in section 2; after that colouring problem has been solved, section 7 completes the proof.

The conjecture was recorded and studied by Fleischner [Fle80]. Arbitrary transition systems need not admit a compatible circuit decomposition; the classical obstruction on the complete graph K_5 is one example [FGZZ19]. The theory of forbidden transitions goes back to Kotzig [Kot68]. Important partial results include the planar case [Fle80, FF90] and the K_5 -minor-free case, where a graph is K_5 -minor-free if the complete graph on five vertices cannot be obtained from it by vertex deletions, edge deletions, and edge contractions [FZ00]. Later work placed these results in a broader structural study of transition systems [FGZZ19].

Fleischner showed that it suffices to consider Eulerian graphs whose vertex degrees all lie in $\{4, 6\}$ [Fle84, Lemma 1, p. 236]. This proof does not use the reduction and handles every even degree directly.

The key step is a four-colouring in which the two edges of every tour transition have different colours and each colour has even degree at every vertex. The proof first derives compatible circuits from this certificate. It then records the passage vertices in cyclic order and reduces the certificate to a colouring lemma for that sequence. The lemma is proved by integrating locally chosen differences and applying an odd-parity selection argument.

The proof has also been formalized in Lean. The formalization is available in the author's GitHub repository <https://github.com/gexahedron/sabidussi-lean>.

Statement of AI use. The proof in this paper is entirely due to GPT 5.6 Pro, and the writeup was prepared with Codex (with GPT 5.6 Sol).

2 From Euler tours to cyclic words

2.1 Encoding the Euler tour

We now make the transition system of an Euler tour explicit, including the incidence bookkeeping needed for loops. With the preceding

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conventions, a loop by itself is a circuit, and two parallel edges together may form a circuit.

Let G and T satisfy the hypotheses of theorem 1.1. Put

$$m = |E(G)| \quad \text{and} \quad I_m = \{0, 1, \dots, m-1\}.$$

Write the edges of T in cyclic order as

$$T = (e_0, e_1, \dots, e_{m-1}),$$

with indices understood modulo m . Temporarily orient each e_i in the direction in which T traverses it, and denote its departure and arrival incidences by d_i and a_i , respectively. Let v_i be the vertex at which the tour passes from e_{i-1} to e_i . Thus a_{i-1} and d_i are incident with v_i , and these two incidences form the transition

$$\tau_i = \{a_{i-1}, d_i\} \quad \text{at } v_i.$$

Under the hypothesis $\delta(G) \geq 4$, one has $m \geq 2$, so the two underlying edges e_{i-1} and e_i in a transition are distinct. Accordingly, when no confusion about incidences is possible, we also write $\tau_i = \{e_{i-1}, e_i\}$.

2.2 The four-colour reduction

Lemma 2.1 (Even decomposition). *The edge set of every finite even multigraph can be partitioned into the edge sets of circuits.*

Proof. Euler's theorem states that every connected finite multigraph containing at least one edge and having only even vertex degrees has a closed trail that contains every edge exactly once. Apply this theorem to a connected component containing an edge. If the resulting trail is not a circuit, some vertex occurs at two distinct positions in its cyclic vertex sequence. Split at two consecutive occurrences of that vertex in the cyclic order to obtain two shorter closed trails. Repeating this operation terminates and leaves closed trails with no repeated vertex apart from the coincident initial and terminal positions. Each is a circuit; a one-edge trail is a loop. Applying the argument to every component containing an edge gives the desired edge partition. $\square$

Let $\mathbb{F}_2 = \{0, 1\}$ be the field with two elements, with arithmetic modulo two, and let

$$\mathcal{W} = \mathbb{F}_2^2.$$

For an edge colouring $\chi : E(G) \rightarrow \mathcal{W}$ and a colour $z \in \mathcal{W}$, define the colour class G_z by

$$V(G_z) = V(G), \quad E(G_z) = \chi^{-1}(z),$$

with all edge incidences inherited from G .

Lemma 2.2 (Four-colour reduction). *Suppose that $\chi : E(G) \rightarrow \mathcal{W}$ has the following two properties:*

1. the two underlying edges of every transition of T have different colours;
2. every colour class G_z , $z \in \mathcal{W}$, is even.

Then G has a circuit decomposition compatible with every transition of T .

Proof. By lemma 2.1, each G_z decomposes into edge-disjoint circuits. The circuits obtained from all four colours partition $E(G)$. Each of them uses edges of only one colour, whereas the two underlying edges in every transition have different colours. Hence no one of these circuits contains a transition. $\square$

2.3 The cyclic-word formulation

It remains to construct the colouring required by lemma 2.2. The edges of the tour will be the gaps in a cyclic word.

Let $n \geq 1$, and let A be a finite set, called the *alphabet*. A *cyclic word* of length n on A is a sequence

$$\mathbf{w} = (w_0, w_1, \dots, w_{n-1})$$

with each $w_i \in A$, considered up to cyclic rotation. We take A to be exactly the set of letters occurring in the word. Put

$$I_n = \{0, 1, \dots, n-1\},$$

and read subscripts modulo n . An index $i \in I_n$ is an *occurrence* of the letter w_i . The word has n gaps, with gap i lying after w_i . The previous and current gap incidences at occurrence i are the ordered pairs

$$(i, \text{previous}) \quad \text{and} \quad (i, \text{current}),$$

associated with gaps $i-1$ and i , respectively. Gap incidences are counted with multiplicity. For $a \in A$, write

$$O_a = \{i \in I_n : w_i = a\}.$$

The sets O_a , $a \in A$, partition I_n .

A colouring of the gaps of a cyclic word by $\mathcal{W}$ is called *admissible* when:

1. the two gaps incident with every occurrence have different colours;
2. for every letter a and every colour $z \in \mathcal{W}$, the number of incidences of colour z at occurrences of a is even.

Returning to the fixed graph G and tour T , for a vertex v let

$$O_T(v) = \{i \in I_m : v_i = v\},$$

and let $H_G(v)$ be the set of edge incidences of G at v . Define

$$\begin{aligned}\Phi_v : O_T(v) \times \{\text{previous, current}\} &\longrightarrow H_G(v), \\ (i, \text{previous}) &\longmapsto a_{i-1}, \\ (i, \text{current}) &\longmapsto d_i.\end{aligned}$$

Lemma 2.3 (Tour-incidence correspondence). *For every $v \in V(G)$, the map Φ_v is a bijection.*

Proof. Every edge incidence belongs to a unique tour edge e_j . If it is the departure incidence d_j , its unique preimage is $(j, \text{current})$; if it is the arrival incidence a_j , its unique preimage is $(j+1, \text{previous})$. The two cases are disjoint, including when e_j is a loop, because a loop still has two distinct incidences. This proves both surjectivity and injectivity. $\square$

The transition pairs at a vertex therefore partition all edge incidences at that vertex. In particular, for every $v \in V(G)$,

$$\deg_G(v) = 2|O_T(v)|. \quad (1)$$

The *tour word* associated with T is

$$\mathbf{w}_T = (v_0, v_1, \dots, v_{m-1}),$$

viewed as a cyclic word on the alphabet $V(G)$.

The following proposition is the promised conversion from the graph problem to this purely combinatorial colouring problem.

Proposition 2.4 (Cyclic-word reduction). *Every admissible colouring of the gaps of $\mathbf{w}_T$ induces an edge colouring $\chi : E(G) \rightarrow \mathcal{W}$ in which the two underlying edges of every transition have different colours and every colour class is even. Consequently, G has a circuit decomposition compatible with every transition of T .*

Proof. Denote the colour of gap i by $x_i \in \mathcal{W}$, and define

$$\chi(e_i) = x_i \quad (i \in I_m).$$

At the transition τ_i , the two underlying tour edges are e_{i-1} and e_i . Their colours are the distinct colours x_{i-1} and x_i of the two gaps incident with occurrence i .

Fix $v \in V(G)$ and $z \in \mathcal{W}$. Under the bijection Φ_v of lemma 2.3, an incidence of G_z at v corresponds exactly to a gap incidence of colour z at an occurrence of v : the previous and current incidences at $i \in O_T(v)$ carry the colours of e_{i-1} and e_i , respectively. Admissibility therefore says that $\deg_{G_z}(v)$ is even. Hence every G_z is even, and lemma 2.2 supplies the required compatible circuit decomposition. $\square$

Every vertex occurs in $\mathbf{w}_T$, because G is connected and has positive minimum degree. Moreover, $O_v = O_T(v)$, and eq. (1) gives

$$|O_v| = \frac{\deg_G(v)}{2} \geq 2 \quad (v \in V(G)).$$

Thus the graph theorem has been reduced to showing that every cyclic word in which each letter occurs at least twice has an admissible gap colouring.

3 The four-colour parity test

Recall that $\mathcal{W} = \mathbb{F}_2^2$, and put

$$\Omega = \mathcal{W} \setminus \{0\}.$$

Thus $\Omega = \{(1,0), (0,1), (1,1)\}$. For $x = (x_1, x_2)$ and $y = (y_1, y_2)$ in $\mathcal{W}$, define the bracket

$$[\cdot, \cdot] : \mathcal{W} \times \mathcal{W} \longrightarrow \mathbb{F}_2$$

and the quadratic form

$$\mathfrak{q} : \mathcal{W} \longrightarrow \mathbb{F}_2$$

by

$$[x, y] = x_1 y_2 + x_2 y_1, \quad (2)$$

$$\mathfrak{q}(x) = x_1 x_2. \quad (3)$$

The bracket is bilinear, symmetric, and alternating, where alternating means that $[x, x] = 0$ for every $x \in \mathcal{W}$. Symmetry and alternation coexist because the ground field has characteristic two. Moreover,

$$\mathfrak{q}(x + y) = \mathfrak{q}(x) + \mathfrak{q}(y) + [x, y]. \quad (4)$$

Iterating eq. (4) gives the following familiar identity.

Lemma 3.1 (Finite polarization). *Let I be a finite linearly ordered set, and let $(z_i)_{i \in I}$ be a family in $\mathcal{W}$. Then*

$$\mathfrak{q} \left(\sum_{i \in I} z_i \right) = \sum_{i \in I} \mathfrak{q}(z_i) + \sum_{\substack{i, j \in I \\ j < i}} [z_i, z_j]. \quad (5)$$

Proof. Induct on $|I|$, adding the largest element last and using eq. (4) and bilinearity of the bracket. $\square$

Let I be a finite set and $a : I \rightarrow \mathcal{W}$. We call the vector $\sum_{i \in I} a(i) \in \mathcal{W}$ its *linear moment*, and the scalar $\sum_{i \in I} \mathfrak{q}(a(i)) \in \mathbb{F}_2$ its *quadratic moment*. We shall recover the parity of all four colour classes from the cardinality of the family and these two moments.

Lemma 3.2 (Four-colour moments). *Let I be a finite set of even cardinality and let $a : I \rightarrow \mathcal{W}$. If*

$$\sum_{i \in I} a(i) = 0 \quad \text{and} \quad \sum_{i \in I} q(a(i)) = 0,$$

then every element of $\mathcal{W}$ occurs an even number of times among the values of a .

Proof. For $z \in \mathcal{W}$, let $n_z \in \mathbb{F}_2$ be the parity of $|\{i \in I : a(i) = z\}|$. For $r, s \in \mathbb{F}_2$, abbreviate $n_{(r,s)}$ to n_{rs} . Since $|I|$ is even,

$$n_{00} + n_{10} + n_{01} + n_{11} = 0.$$

The two coordinates of $\sum_{i \in I} a(i) = 0$ give

$$n_{10} + n_{11} = 0, \quad n_{01} + n_{11} = 0.$$

Finally, q is one precisely at $(1, 1)$, so the quadratic moment gives $n_{11} = 0$. The four displayed equations imply $n_{00} = n_{10} = n_{01} = n_{11} = 0$. $\square$

4 The odd three-state balancing lemma

Let U be a finite set and let Σ be a three-element set, whose elements we call *states*. For each ordered pair (u, v) of distinct elements of U , let

$$b_{uv} : \Sigma \times \Sigma \longrightarrow \mathbb{F}_2.$$

Write Σ^U for the set of functions $\sigma : U \rightarrow \Sigma$, and put $\sigma_u = \sigma(u)$. For $\sigma \in \Sigma^U$, define

$$Q_u(\sigma) = \sum_{\substack{v \in U \\ v \neq u}} b_{uv}(\sigma_u, \sigma_v) \quad (u \in U).$$

Call σ *balanced* when $Q_u(\sigma) = 0$ for every $u \in U$. For $S \subseteq U$, put

$$Z(S) = \sum_{\sigma \in \Sigma^U} \prod_{u \in S} Q_u(\sigma),$$

where an empty product equals 1.

The next lemma shows that states can be chosen simultaneously so that all balance equations hold. It uses only the symmetry and row-sum hypotheses, not bilinearity.

Lemma 4.1 (Odd balancing). *Suppose that the maps b_{uv} satisfy*

$$b_{uv}(r, s) = b_{vu}(s, r) \quad (r, s \in \Sigma), \quad (6)$$

$$\sum_{s \in \Sigma} b_{uv}(r, s) = 0 \quad (r \in \Sigma). \quad (7)$$

Then the number of balanced assignments in Σ^U is odd. In particular, a balanced assignment exists.

Proof. All algebraic calculations in this proof are in $\mathbb{F}_2$. Let N be the integer number of balanced assignments, and let $\bar{N} \in \mathbb{F}_2$ be its parity. Since $1 + \alpha$ is the indicator of $\alpha = 0$ for $\alpha \in \mathbb{F}_2$, expanding the product gives

$$\bar{N} = \sum_{\sigma \in \Sigma^U} \prod_{u \in U} (1 + Q_u(\sigma)) = \sum_{S \subseteq U} Z(S). \quad (8)$$

The empty term is

$$Z(\emptyset) = \sum_{\sigma \in \Sigma^U} 1 = 3^{|U|} = 1$$

in $\mathbb{F}_2$, because $3 = 1$ in this field. We show that $Z(S) = 0$ whenever $S \neq \emptyset$.

For every map $f : S \rightarrow U$ satisfying $f(u) \neq u$ for all $u \in S$, put

$$Z_f = \sum_{\sigma \in \Sigma^U} \prod_{u \in S} b_{uf(u)}(\sigma_u, \sigma_{f(u)}).$$

Expanding one summand from each Q_u , $u \in S$, gives

$$Z(S) = \sum_{\substack{f: S \rightarrow U \\ f(u) \neq u \text{ for all } u \in S}} Z_f. \quad (9)$$

Let H_f be the directed graph on vertex set U having one arc $u \rightarrow f(u)$ for each $u \in S$. For $w \in U$, let $\deg_f(w)$ count the arc ends incident with w , disregarding orientation but retaining arc multiplicity.

Suppose first that H_f has a vertex w with $\deg_f(w) = 1$. The variable σ_w occurs in exactly one factor of the product defining Z_f . After all other variables are fixed, summing over $\sigma_w \in \Sigma$ gives zero by eq. (7); if w is the source rather than the target of its unique arc, first use eq. (6). Hence $Z_f = 0$.

Now suppose that H_f has no vertex of degree one. Let

$$U_f = S \cup \{f(u) : u \in S\}$$

be the set of vertices incident with an arc. There are $|S|$ arcs, so

$$2|S| = \sum_{w \in U_f} \deg_f(w) \geq 2|U_f| \geq 2|S|.$$

Equality holds throughout. Therefore $U_f = S$, every vertex of S has degree two, and f is a fixed-point-free permutation of S .

The variables indexed by $U \setminus S$ occur in no factor and contribute $3^{|U \setminus S|} = 1$ in $\mathbb{F}_2$. Because f is a permutation, the remaining product and assignment sum factor over the disjoint cycles of f . If f contains the two-cycle $u \leftrightarrow v$, the factor belonging to this cycle is

$$\begin{aligned} \sum_{r, s \in \Sigma} b_{uv}(r, s) b_{vu}(s, r) &= \sum_{r, s \in \Sigma} b_{uv}(r, s)^2 \\ &= \sum_{r, s \in \Sigma} b_{uv}(r, s) = 0, \end{aligned}$$

where we used symmetry, the identity $\alpha^2 = \alpha$ for every $\alpha \in \mathbb{F}_2$, and the row-sum condition. Thus $Z_f = 0$ whenever f has a two-cycle.

It remains to consider fixed-point-free permutations all of whose cycles have length at least three. Such an f is distinct from f^{-1} . Symmetry shows, factor by factor around every permutation cycle, that

$$Z_f = Z_{f^{-1}}.$$

The remaining permutations therefore cancel in inverse pairs. This proves $Z(S) = 0$ for every nonempty S . Equation (8) now gives $\overline{N} = 1$, so N is odd. $\square$

5 Local difference patterns

From now on, the state set is $\Omega = \mathcal{W} \setminus \{0\}$. Define the linear automorphism

$$\rho : \mathcal{W} \longrightarrow \mathcal{W}, \quad \rho(r, s) = (s, r + s) \quad ((r, s) \in \mathcal{W}).$$

It has order three and cyclically permutes the three elements of Ω :

$$(1, 0) \longmapsto (0, 1) \longmapsto (1, 1) \longmapsto (1, 0).$$

In particular,

$$\sum_{t \in \Omega} t = 0. \tag{10}$$

Let O be a finite linearly ordered set, and put $k = |O| \geq 2$. For each state $t \in \Omega$, define a difference pattern

$$D_{O,t} : O \longrightarrow \Omega$$

as follows:

- If k is even, set $D_{O,t}(i) = t$ for every $i \in O$.
- If k is odd, write $O = \{i_1 < \cdots < i_k\}$, and put

$$\begin{aligned} D_{O,t}(i_j) &= t & (1 \leq j \leq k-2), \\ D_{O,t}(i_{k-1}) &= \rho t, & D_{O,t}(i_k) &= \rho^2 t. \end{aligned}$$

For odd k , this is obtained from the basic terminal triple

$$(t, \rho t, \rho^2 t)$$

by prepending pairs (t, t) recursively. Thus the construction pairs off all occurrences except for one terminal triple. The following properties are immediate but decisive.

Lemma 5.1 (Local patterns). *Let O be a finite linearly ordered set with $|O| \geq 2$. Then*

$$D_{O,t}(i) \neq 0 \quad (t \in \Omega, i \in O), \quad (11)$$

$$\sum_{i \in O} D_{O,t}(i) = 0 \quad (t \in \Omega), \quad (12)$$

$$\sum_{t \in \Omega} D_{O,t}(i) = 0 \quad (i \in O). \quad (13)$$

Proof. Nonvanishing follows because $t, \rho t, \rho^2 t$ lie in Ω . If k is even, the sum in eq. (12) consists of an even number of copies of t . If k is odd, the first $k - 3$ values form an even number of pairs of t , while the final three values $t, \rho t, \rho^2 t$ are all three nonzero elements of $\mathcal{W}$ and hence sum to zero.

At any fixed $i \in O$, the map $t \mapsto D_{O,t}(i)$ is one of the three maps $t \mapsto t$, $t \mapsto \rho t$, and $t \mapsto \rho^2 t$. It therefore permutes Ω , and eq. (13) follows from eq. (10). $\square$

6 Construction of the gap colouring

Fix a cyclic word $\mathbf{w}$ for which $|O_a| \geq 2$ for every $a \in A$. Choose the representative $\mathbf{w} = (w_0, \dots, w_{n-1})$ and cut its cyclic order immediately before w_0 . Thus

$$0 < 1 < \dots < n - 1,$$

and every occurrence set O_a , $a \in A$, inherits a linear order. Suppose for a moment that the desired gap colours are $x_0, \dots, x_{n-1} \in \mathcal{W}$, and put

$$y_i = x_{i-1} + x_i \quad (i \in I_n).$$

The two gaps at occurrence i have different colours exactly when $y_i \neq 0$. Conversely, nonzero differences y_i whose sum is zero can be integrated around the cyclic word by taking prefix sums. We shall choose the differences so that the stronger local equations

$$\sum_{i \in O_a} y_i = 0 \quad (a \in A)$$

hold. These equations imply the global zero sum and will also give the linear part of the required incidence parity. The remaining, quadratic part is encoded by the following four-colour test.

6.1 The difference formulation

For a family $y = (y_i)_{i \in I_n}$ in $\mathcal{W}$ and a letter $a \in A$, define its *external quadratic contribution* at a by

$$R_a(y) = \sum_{\substack{b \in A \\ b \neq a}} \sum_{\substack{i \in O_a, j \in O_b \\ j < i}} [y_i, y_j]. \quad (14)$$

The next lemma isolates exactly what the difference construction must achieve.

Lemma 6.1 (Difference criterion). *Suppose that $y_i \in \mathcal{W}$ for $i \in I_n$, and that*

$$y_i \neq 0 \quad (i \in I_n), \quad (15)$$

$$\sum_{i \in O_a} y_i = 0 \quad (a \in A), \quad (16)$$

$$R_a(y) = 0 \quad (a \in A). \quad (17)$$

Then the gaps of $\mathbf{w}$ have an admissible colouring.

Proof. The occurrence sets partition I_n , so eq. (16) gives

$$\sum_{i=0}^{n-1} y_i = 0. \quad (18)$$

Set $x_{-1} = 0$ and, with respect to the chosen cut, define

$$x_i = \sum_{j=0}^i y_j \quad (0 \leq i < n).$$

By eq. (18), $x_{n-1} = 0 = x_{-1}$, so these prefix sums close cyclically. Read the subscripts on x modulo n , and colour gap i by x_i . For every $i \in I_n$,

$$x_{i-1} + x_i = y_i \neq 0, \quad (19)$$

where the case $i = 0$ uses $x_{n-1} = x_{-1} = 0$. Thus the two gaps at every occurrence have different colours.

Fix $a \in A$, and let

$$J_a = O_a \times \{\text{previous, current}\}$$

be the set of gap incidences at occurrences of a . It has even cardinality $2|O_a|$, and its indexed family of colours is

$$(x_{i-1}, x_i)_{i \in O_a}.$$

The linear moment of this family is

$$\sum_{i \in O_a} (x_{i-1} + x_i) = \sum_{i \in O_a} y_i = 0. \quad (20)$$

For $i \in O_a$, eqs. (4) and (19) give

$$\mathbf{q}(x_{i-1}) + \mathbf{q}(x_i) = \mathbf{q}(y_i) + [y_i, x_{i-1}]. \quad (21)$$

The strict-prefix identity

$$x_{i-1} = \sum_{0 \leq j < i} y_j$$

also holds for $i = 0$, when both sides are zero. Splitting its indices according to the partition $(O_b)_{b \in A}$, we obtain

$$\begin{aligned} P_a &:= \sum_{i \in O_a} (\mathfrak{q}(x_{i-1}) + \mathfrak{q}(x_i)) \\ &= \sum_{i \in O_a} \mathfrak{q}(y_i) + \sum_{\substack{i, j \in O_a \\ j < i}} [y_i, y_j] + R_a(y). \end{aligned} \quad (22)$$

By lemma 3.1 and eq. (16), the first two terms on the last line equal

$$\mathfrak{q} \left(\sum_{i \in O_a} y_i \right) = 0.$$

The last term is zero by eq. (17); hence $P_a = 0$. The family indexed by J_a has even cardinality, zero linear moment by eq. (20), and zero quadratic moment by $P_a = 0$. By lemma 3.2, every colour occurs an even number of times in this family. Since a was arbitrary, every colour has even incidence at every letter, so the colouring is admissible. $\square$

6.2 Interactions between letters

Given a state $t_a \in \Omega$ for each letter a , set

$$y_i = D_{O_{w_i}, t_{w_i}}(i) \quad (i \in I_n).$$

By lemma 5.1, these differences are nonzero and have zero sum on every occurrence set. It remains only to make their external quadratic contributions vanish.

For distinct letters $v, u \in A$, define their interaction

$$b_{vu} : \Omega \times \Omega \longrightarrow \mathbb{F}_2$$

by

$$b_{vu}(t, s) = \sum_{\substack{i \in O_v, j \in O_u \\ j < i}} [D_{O_v, t}(i), D_{O_u, s}(j)] \quad (t, s \in \Omega). \quad (23)$$

For any choice of the states, the definition of R_v gives

$$R_v(y) = \sum_{\substack{u \in A \\ u \neq v}} b_{vu}(t_v, t_u) \quad (v \in A). \quad (24)$$

Lemma 6.2 (Interaction tables). *For distinct letters $v, u \in A$, the interactions satisfy*

$$b_{vu}(t, s) = b_{uv}(s, t) \quad (t, s \in \Omega), \quad (25)$$

$$\sum_{s \in \Omega} b_{vu}(t, s) = 0 \quad (t \in \Omega). \quad (26)$$

Proof. Fix $t, s \in \Omega$. Since $v \neq u$, positions $i \in O_v$ and $j \in O_u$ are distinct, so exactly one of $j < i$ and $i < j$ holds. Using symmetry and bilinearity of the bracket,

$$\begin{aligned} b_{vu}(t, s) + b_{uv}(s, t) &= \sum_{i \in O_v} \sum_{j \in O_u} [D_{O_v, t}(i), D_{O_u, s}(j)] \\ &= \left[\sum_{i \in O_v} D_{O_v, t}(i), \sum_{j \in O_u} D_{O_u, s}(j) \right] = 0. \end{aligned} \quad (27)$$

The last equality follows from eq. (12). Since the codomain is $\mathbb{F}_2$, it proves eq. (25). For a fixed t ,

$$\sum_{s \in \Omega} b_{vu}(t, s) = \sum_{\substack{i \in O_v, j \in O_u \\ j < i}} \left[D_{O_v, t}(i), \sum_{s \in \Omega} D_{O_u, s}(j) \right] = 0, \quad (28)$$

by eq. (13). This proves eq. (26). $\square$

The interaction tables therefore satisfy the hypotheses of lemma 4.1.

6.3 Completion of the gap colouring

Lemma 6.3 (Gap colouring). *Suppose that $|O_a| \geq 2$ for every $a \in A$. The gaps of $\mathbf{w}$ can be coloured by $\mathcal{W}$ so that:*

1. *the two gaps incident with every occurrence have different colours;*
2. *for every letter a and every colour $z \in \mathcal{W}$, the number of incidences of colour z at occurrences of a is even.*

Proof. By lemma 6.2, the concrete maps b_{vu} satisfy the hypotheses of lemma 4.1. Apply that lemma with $U = A$ and $\Sigma = \Omega$. It supplies a state $t_a \in \Omega$ for each $a \in A$ such that

$$\sum_{\substack{b \in A \\ b \neq a}} b_{ab}(t_a, t_b) = 0 \quad (a \in A). \quad (29)$$

For $i \in I_n$, put

$$y_i = D_{O_{w_i}, t_{w_i}}(i).$$

By eqs. (11) and (12),

$$y_i \neq 0 \quad (i \in I_n), \quad \sum_{i \in O_a} y_i = 0 \quad (a \in A).$$

Finally, eqs. (24) and (29) give

$$R_a(y) = 0 \quad (a \in A).$$

All three hypotheses of lemma 6.1 hold, so that lemma supplies the required gap colouring. $\square$

Remark 6.4. Once the occurrence orders are fixed, lemma 4.1 shows that the number of choices of local patterns satisfying all external balance equations is odd.

7 Proof of the graph theorem

Theorem (Sabidussi; restatement of theorem 1.1). *Let G be a finite Eulerian multigraph with $\delta(G) \geq 4$, and let T be an Euler tour of G . There is a circuit decomposition $\mathcal{C}$ of G such that no $C \in \mathcal{C}$ contains both underlying edges of any transition of T .*

Proof of theorem 1.1. For the tour word $\mathbf{w}_T$, the calculation at the end of section 2 gives $|O_v| = \deg_G(v)/2 \geq 2$ for every $v \in V(G)$. By lemma 6.3, this word has an admissible gap colouring. The cyclic-word reduction, proposition 2.4, now supplies the required compatible circuit decomposition. $\square$

Corollary 7.1 (Four-colour certificate). *Under the hypotheses of theorem 1.1, there is an edge colouring*

$$\chi : E(G) \longrightarrow \mathbb{F}_2^2$$

such that the two underlying edges of every transition of T have different colours and every colour class is an even subgraph.

Proof. This is the edge colouring supplied by proposition 2.4 when the admissible colouring from lemma 6.3 is applied to the tour word. $\square$

7.1 Some additional remarks

The minimum-degree hypothesis is visible at the local-pattern stage. A letter must occur at least twice, equivalently a vertex of the Eulerian graph must have degree at least four. Degree two is a genuine obstruction: at such a vertex the sole transition consists of its two incident edge-ends, and every circuit through the vertex contains all underlying edges represented by that transition.

The special cyclic order supplied by one Euler tour enters the algebra in eq. (27). The chosen cut linearly orders all occurrences, and the two possible orders of a pair of distinct letters exhaust their Cartesian product. An arbitrary transition system need not arise from a single cyclic edge order in which the paired incidences are consecutive, so the same symmetry need not be available.

A colour class may have degree greater than two at a vertex whose degree in G is greater than six. Evenness of every colour class is the invariant delivered by the algebra. The opening graph-theoretic reduction separates that invariant from the elementary decomposition of an even subgraph into circuits; this is what allows the construction to handle every even degree directly.

8 Circuit double covers and dominating circuits

This section contains no new material. It is included only to remind the reader where Sabidussi's compatibility conjecture fits into the broader circuit-double-cover picture and how it relates to the other conjectures discussed below.

Recall that every circuit in this paper is connected.

For this discussion, a graph is *simple* when it has neither loops nor parallel edges, and it is *cubic* when every vertex has degree three. A *bridge* is an edge whose deletion increases the number of connected components, and a graph is *bridgeless* when it has no bridge. A *circuit double cover*, abbreviated *CDC*, of a graph X is a finite family of circuits of X , with repetitions allowed, in which every edge belongs to exactly two members, counted with multiplicity. In the standard bounded-cover terminology, a *cycle* is an even subgraph and need not be connected; a *k-cycle double cover*, or *k-CDC*, is a family of at most k such cycles in which every edge belongs to exactly two members. The circuit double cover conjecture asserted that every finite bridgeless graph with at least one edge has a CDC. OpenAI recently proved the stronger 8-CDC theorem [Ope26]. Indeed, its $\Gamma = \mathbb{F}_2^3$ construction produces the eight even subgraphs M_s , $s \in \Gamma$, with every edge in exactly two of them. Decomposing the M_s into connected circuit components gives a CDC in the convention used here.

The *strong circuit double cover conjecture*, also called the fixed-circuit conjecture, asserts that for every finite bridgeless cubic graph X and every specified circuit C of X , there is a CDC of X containing C [Fle84, God88, Zha12]. A circuit C is *dominating* when every edge of X has at least one end in $V(C)$. An equivalent cubic formulation of Sabidussi compatibility states that whenever a finite cubic graph X has a dominating circuit C , the graph X has a CDC containing C [Zha12, FGK24]. Consequently, theorem 1.1 proves the strong-CDC conclusion whenever the specified circuit is dominating, but not for an arbitrary specified circuit.

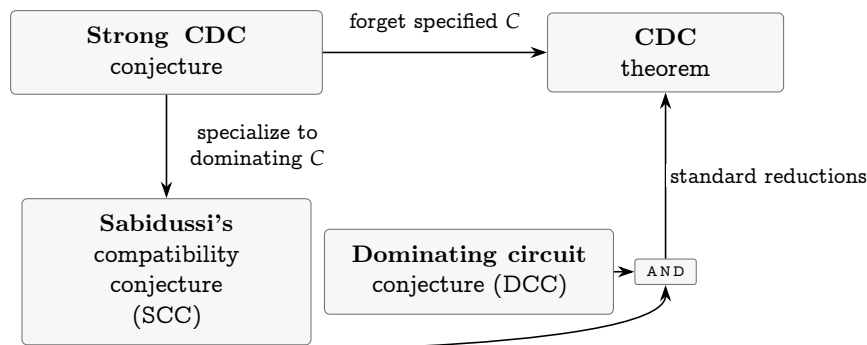
To state the implication for the ordinary CDC conjecture, an *edge cut* is a set of edges whose deletion disconnects a connected graph. Such a cut is *cyclic* when at least two of the resulting components contain a circuit. A connected graph is *cyclically 4-edge-connected* when every cyclic edge cut has at least four edges. A *proper 3-edge-colouring* assigns three colours to the edges so that edges sharing an end receive different colours. Fleischner's original *Dominating Circuit Conjecture* asserts that every finite simple cyclically 4-edge-connected cubic graph either has a proper 3-edge-colouring or has a dominating circuit [Fle84]. Equivalently, every finite simple cyclically 4-edge-connected cubic graph without a proper 3-edge-colouring has a dominating circuit [BFK⁺08]. A stronger commonly used formulation

OpenAI. A proof of the cycle double cover conjecture. Preprint, July 2026. https://cdn.openai.com/pdf/04d1d1e4-bc75-476a-97cf-49055cd98d31/cdc_proof.pdf

Herbert Fleischner. Cycle decompositions, 2-coverings, removable cycles and the four-color disease. In J. A. Bondy and U. S. R. Murty, editors, *Progress in Graph Theory*, pages 233–246. Academic Press, New York, 1984; Luis Armando Goddyn. *Cycle Covers of Graphs*. PhD thesis, University of Waterloo, Waterloo, Ontario, Canada, 1988; and Cun-Quan Zhang. *Circuit Double Cover of Graphs*, volume 399 of *London Mathematical Society Lecture Note Series*. Cambridge University Press, Cambridge, 2012. <https://doi.org/10.1017/CB09780511863158> Herbert Fleischner, Behrooz Bagheri Gh., and Benedikt Klocker. Perfect pseudo-matchings in cubic graphs. *Graphs and Combinatorics*, 40:118, 2024. <https://doi.org/10.1007/s00373-024-02844-y>

requires a dominating circuit in every finite simple cyclically 4-edge-connected cubic graph [FK02].

Historically, together with the standard reductions for the circuit double cover conjecture and the known CDC construction for properly 3-edge-colourable cubic graphs [Jae85, Zha12], this gives the literature implication that Sabidussi compatibility plus Fleischner’s original form—and hence also plus the stronger Dominating Circuit Conjecture—implies the circuit double cover conjecture. The relevant dominating-circuit statements remain open, so this remains a conditional route to the CDC theorem. The logical dependencies are summarized in fig. 1; the lower route uses SCC and DCC jointly, while OpenAI’s 8-CDC theorem establishes CDC independently of either implication route.



Herbert Fleischner and Martin Kochol. A note about the dominating circuit conjecture. *Discrete Mathematics*, 259(1–3):307–309, 2002. [https://doi.org/10.1016/S0012-365X\(02\)00588-5](https://doi.org/10.1016/S0012-365X(02)00588-5)

Figure 1: Logical dependencies among the four statements.

9 A dimer and spin interpretation

The algebra used in the gap-colouring proof has a direct genus-one interpretation. This interpretation is not needed for the proof, but it explains why the bracket, the quadratic moment, and the strict-prefix sums fit together so naturally. Choose a symplectic basis of the first homology of the torus and identify

$$\mathcal{W} = H_1(T^2; \mathbb{F}_2).$$

Under this identification, the bracket in eq. (2) is the mod-two intersection pairing on the torus. The polarization identity eq. (4) says exactly that q is a quadratic refinement of that pairing. Such refinements represent spin structures on the torus. More generally, a Kasteleyn orientation together with a reference perfect matching of a graph embedded in an oriented surface determines a quadratic refinement of the surface intersection form [CR07].

9.1 The four-matchings dimer

Let Δ_4 be the toroidal four-dipole: one black vertex and one white vertex joined by four edges in the standard cellular embedding with

David Cimasoni and Nicolai Reshetikhin. Dimers on surface graphs and spin structures. I. *Communications in Mathematical Physics*, 275(1):187–208, 2007. <https://doi.org/10.1007/s00220-007-0302-7>

two quadrilateral faces. It is the dimer graph usually associated with the conifold [HK05]. Each edge by itself is a perfect matching. Label these matchings D_x , $x \in \mathcal{W}$, so that, after fixing D_0 ,

$$[D_x \triangle D_0] = x \quad \text{in } H_1(T^2; \mathbb{F}_2).$$

Here and below, the symmetric difference of two matchings is regarded as a mod-two cycle.

There is a Kasteleyn orientation K_0 for which the normalized characteristic polynomial is

$$P(z, w) = 1 + z + w - zw = \sum_{x \in \mathcal{W}} (-1)^{q(x)} z^{x_1} w^{x_2}. \quad (30)$$

The variables record the two homology coordinates. A common conifold convention gives $1 - z - w - zw$; twisting the Kasteleyn orientation along both fundamental cycles, equivalently replacing (z, w) by $(-z, -w)$, gives eq. (30).

Write q_D^K for the quadratic refinement associated with a Kasteleyn orientation K and a reference matching D . We regard its Arf invariant as an element of $\mathbb{F}_2$.

Proposition 9.1 (Conifold sign dictionary). *With the preceding normalization,*

$$q_{D_0}^{K_0}(y) = q(y), \quad (31)$$

$$q_{D_x}^{K_0}(y) = q(y) + [x, y], \quad (32)$$

$$\text{Arf}(q_{D_x}^{K_0}) = q(x) \quad (33)$$

for all $x, y \in \mathcal{W}$. In particular, if a matching in sector x is switched to one in sector $x + y$, then

$$q_{D_x}^{K_0}(y) = q(x) + q(x + y). \quad (34)$$

Proof. After the coefficient of D_0 is normalized to be positive, eq. (30) says that the relative Kasteleyn sign of D_x is $(-1)^{q(x)}$. This is eq. (31). Changing the reference matching from D_0 to D_x translates the associated quadratic refinement by intersection with $[D_x \triangle D_0] = x$, which gives eq. (32). If $x = (a, b)$ and e_1, e_2 is the chosen symplectic basis, then

$$q_{D_x}^{K_0}(e_1) = b, \quad q_{D_x}^{K_0}(e_2) = a.$$

The genus-one definition of the Arf invariant therefore gives

$$\text{Arf}(q_{D_x}^{K_0}) = ab = q(x).$$

This proves eq. (33). Finally, eq. (34) follows from eqs. (4) and (32). $\square$

At an occurrence i in the word, take $x = x_{i-1}$ and $y = y_i = x_{i-1} + x_i$. Then $[D_{x_{i-1}} \triangle D_{x_i}] = y_i$, and eq. (34) is precisely the quadratic-pair identity eq. (21). Thus the bit occurring there is the exponent of the relative Kasteleyn sign for the switch between the two gap matchings.

The four-colour moment lemma has an equally literal spin interpretation. By eq. (33), the linear moment of a family of colours is its total matching-sector homology, while its quadratic moment is the total Arf parity of the induced spin structures. Hence lemma 3.2 says that an even family of the four torus matching sectors, with zero total homology and zero total Arf parity, contains every sector an even number of times. In coordinates, this is the fact that the four functions

$$1, \quad x_1, \quad x_2, \quad x_1x_2$$

form a basis of all $\mathbb{F}_2$ -valued functions on $\mathcal{W}$.

Deleting the edge corresponding to D_0 from Δ_4 gives the toroidal three-dipole. Its three perfect matchings are indexed by $\Omega = \mathcal{W} \setminus \{0\}$. At the level of matching sectors, the order-three symplectic change of homology basis acts as

$$\rho(r, s) = (s, r + s).$$

This is the three-state atom used in section 5. There is also a spin-only description of these states. The odd quadratic refinement

$$q_*(x) = x_1 + x_2 + x_1x_2$$

is ρ -invariant, and $t \mapsto q_* + [t, \cdot]$ identifies Ω , ρ -equivariantly, with the three even torus spin structures. Since $q_* - \mathfrak{q}$ is linear, replacing $\mathfrak{q}$ by q_* does not change any quadratic moment in the proof for which the corresponding linear moment vanishes.

9.2 The prefix order as linking

The strict-prefix condition in eq. (14) also has a topological realization. Place the word positions at distinct points $p_0, \dots, p_{n-1}$, in this cyclic order, on S^1 , and put

$$M = S^1 \times T^2.$$

For each $y \in \mathcal{W}$, choose a mod-two cycle γ_y on the torus representing y . Given a family $(y_i)_{i \in I_n}$ satisfying the local equations $\sum_{i \in O_a} y_i = 0$, define

$$C_a = \sum_{i \in O_a} \{p_i\} \times \gamma_{y_i} \quad \text{in } M. \quad (35)$$

Its homology class is

$$[C_a] = \left(0, \sum_{i \in O_a} y_i \right) = 0,$$

so C_a bounds a mod-two two-chain. If C and C' are disjoint null-homologous one-cycles in M , let $\text{lk}_2(C, C')$ denote their mod-two linking number. It is the intersection of C' with a two-chain bounded by C ; the null-homology of C' makes this independent of the bounding chain.

Proposition 9.2 (Prefix-linking dictionary). *For distinct letters a, b ,*

$$\text{lk}_2(C_a, C_b) = \sum_{\substack{i \in O_a, j \in O_b \\ j < i}} [y_i, y_j]. \quad (36)$$

Consequently,

$$R_a(y) = \text{lk}_2 \left(C_a, \sum_{b \neq a} C_b \right). \quad (37)$$

Proof. Cut S^1 immediately before p_0 . For each $i \in O_a$, join the cut to p_i by the positively oriented interval I_i . The cylinders $I_i \times \gamma_{y_i}$ have boundary C_a , together with a cycle in the cut fibre representing $\sum_{i \in O_a} y_i = 0$. Cap that residual cycle inside the cut fibre. The cap is disjoint from every C_b .

The cylinder $I_i \times \gamma_{y_i}$ meets $\{p_j\} \times \gamma_{y_j}$ exactly when $j < i$, and its mod-two intersection number then equals the torus intersection $[y_i, y_j]$. Summing these intersections proves eq. (36); summing over $b \neq a$ proves eq. (37). $\square$

For the prefix-integrated colours in the proof, proposition 9.1 and eq. (22) and the local equation $\sum_{i \in O_a} y_i = 0$ give

$$\begin{aligned} \prod_{i \in O_a} (-1)^{q_{Dx_{i-1}}^{K_0}(y_i)} &= (-1)^{P_a} \\ &= (-1)^{R_a(y)} \\ &= (-1)^{\text{lk}_2(C_a, \sum_{b \neq a} C_b)}. \end{aligned} \quad (38)$$

Thus the external obstruction is the residual Kasteleyn sign after the switches belonging to a letter close homologically, and the chosen prefix order computes that sign as genuine three-dimensional linking.

9.3 The odd-balancing determinant

The global parity argument in lemma 4.1 has a determinantal form closely related to matrix-valued dimer and web-trace expansions. For distinct $u, v \in U$, let

$$B_{uv} = (b_{uv}(r, s))_{r, s \in \Sigma'}$$

put $B_{uu} = 0$, and let $C = (B_{uv})_{u, v \in U}$ be the resulting $3|U|$ by $3|U|$ block matrix. Work over the square-zero activity ring

$$\mathcal{R} = \mathbb{F}_2[x_u \mid u \in U] / (x_u^2 \mid u \in U), \quad X = \text{diag}(x_u I_3 \mid u \in U).$$

Recall that, for $S \subseteq U$,

$$Z(S) = \sum_{\sigma \in \Sigma^U} \prod_{u \in S} Q_u(\sigma).$$

Proposition 9.3 (Balancing determinant). *In $\mathcal{R}$,*

$$\det(I_{3|U|} + XC) = \sum_{S \subseteq U} Z(S) \prod_{u \in S} x_u = 1, \quad (39)$$

Proof. Expand the determinant by principal minors. Since $x_u^2 = 0$, a surviving minor chooses at most one state coordinate over each u . The coefficient of $\prod_{u \in S} x_u$ is therefore

$$\sum_{\sigma: S \rightarrow \Sigma} \det(b_{uv}(\sigma_u, \sigma_v))_{u,v \in S'} \quad (40)$$

with zero diagonal. Expanding this determinant gives the terms indexed by fixed-point-free permutations of S . The leaf-elimination argument in the proof of lemma 4.1 shows that these are exactly the terms remaining in $Z(S)$; variables outside S contribute $3^{|U \setminus S|} = 1$ in $\mathbb{F}_2$. This proves the first equality in eq. (39).

For completeness, the cycle expansion also proves the second equality directly. A permutation cycle $(u_1 u_2 \dots u_k)$ contributes

$$\text{Tr}(B_{u_1 u_2} B_{u_2 u_3} \cdots B_{u_k u_1}).$$

A two-cycle has total contribution

$$\sum_{r,s \in \Sigma} b_{uv}(r,s) b_{vu}(s,r) = \sum_{r,s \in \Sigma} b_{uv}(r,s) = 0$$

by symmetry and the row-sum condition. A permutation with no two-cycle pairs with its inverse; their contributions agree by symmetry and cancel in characteristic two. Thus every nonconstant coefficient is zero. $\square$

There is an effective rank-two state space hidden in this determinant. Let $\mathbf{1} \in \mathbb{F}_2^\Sigma$ be the constant vector and put

$$E = \left\{ v \in \mathbb{F}_2^\Sigma : \sum_{r \in \Sigma} v_r = 0 \right\}.$$

The row-sum condition gives $B_{uv} \mathbf{1} = 0$, and symmetry gives $\mathbf{1}^T B_{uv} = 0$; hence every block preserves the two-dimensional space E . Since $|\Sigma| = 3$, $\mathbb{F}_2^\Sigma = \langle \mathbf{1} \rangle \oplus E$, so eq. (39) reduces to a determinant on E^U , of dimension $2|U|$. Its cycle weights are traces of products of the induced two-dimensional edge maps. This is the same determinant-and-trace mechanism used for dimers with matrix-valued connections and for higher-rank dimer models [DKS24, KO25].

Daniel C. Douglas,
Richard Kenyon, and
Haolin Shi. Dimers,
webs, and local sys-
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377(2):921–950, 2024.
[https://doi.org/10.
.1090/tran/8993](https://doi.org/10.1090/tran/8993); and
Richard Kenyon and
Nicholas Ovenhouse.
Higher-rank dimer
models. *Transac-*

The comparison has a real boundary. The local conifold matching sectors and their Kasteleyn signs are ordinary torus-dimer objects, but eq. (39) lives over a characteristic-two ring with nilpotent activities. The relations $x_\mu^2 = 0$ enforce the choice of at most one state channel at a letter. Accordingly, the determinant should be read as an exact rank-two fermionic or web-trace reformulation of the global parity proof, not as an asserted positive-weight scalar dimer model for the entire certificate construction. The combined picture is a conifold-dimer spin transport whose pairwise obstruction is mod-two linking, completed by the odd three-state balancing lemma.

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